In Lab Tasks



Assignment Title: In Lab 7 Tasks
Subject Name: Programming Fundamentals Lab
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Question 1: Shopkeeper Billing System (2D Array)

A shopkeeper sells 5 different products. Write a C program using a 2D array to store the quantity and unit price of each product. Calculate the total price of each product using quantity × price, display the quantity, unit price, and total price of each product, and calculate and display the grand total of all products using nested loops.

main.c

Run

```c
#include <stdio.h>
int main(){
    int product[5][2];
    int i, j, total, grandTotal = 0;

    for (i = 0; i < 5; i++){
        printf("Enter quantity and price of product %d: ", i + 1);

        for (j = 0; j < 2; j++) {
            scanf("%d", &product[i][j]);
        }
    }
    printf("\nProduct\tQuantity\tPrice\tTotal\n");

    for (i = 0; i < 5; i++){
        total = product[i][0] * product[i][1];

        printf("%d\t%d\t\t%d\t%d\n", i + 1, product[i][0], product[i][1], total);

        grandTotal = grandTotal + total;
    }
    printf("\nGrand Total = %d", grandTotal);

    return 0;
}
```

```
Output

Enter quantity and price of product 1: 2     769
Enter quantity and price of product 2: 2     760
Enter quantity and price of product 3: 4     987
Enter quantity and price of product 4: 3     659
Enter quantity and price of product 5: 5     1390

Product Quantity    Price   Total
1   2       769 1538
2   2       760 1520
3   4       987 3948
4   3       659 1977
5   5       1390    6950

Grand Total = 15933
```

Question 2: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

```
1
1 4
1 4 27
1 4 27 16
1 4 27 16 125
```

main.c

Run ▶

```c
#include <stdio.h>
int main(){
    int i, j;

    for (i = 1; i <= 5; i++){
        for (j = 1; j <= i; j++) {
            if (j % 2 == 0)
                printf("%d ", j * j);
            else
                printf("%d ", j * j * j);
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
1
1 4
1 4 27
1 4 27 16
1 4 27 16 125
```

Question 3: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

main.c Run ▶

```c
#include <stdio.h>
int main(){
    int i, j;

    for (i = 1; i <= 5; i++){
        for (j = 1; j <= i; j++){
            printf("%d ", i);
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

Question 4: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

main.c Run

```c
#include <stdio.h>
int main(){
    int i, j;

    for (i = 5; i >= 1; i--) {
        for (j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Question 5: 3×3 Matrix Types Using 2D Arrays in C

Write a C program that takes input for a 3×3 matrix using a 2D array. Apply appropriate conditions using nested loops to check and display whether the matrix satisfies the following types:

1. Square Matrix
2. Null Matrix
3. Diagonal Matrix
4. Upper Triangular Matrix
5. Lower Triangular Matrix
6. Identity Matrix

Display all applicable matrix types. A single matrix may satisfy multiple conditions.

At the end, using loops, also display the diagonal elements, the upper triangular elements and the lower triangular elements of the matrix.

main.c

Run

```c
#include <stdio.h>
int main(){
    int a[3][3];
    int i, j;
    int null = 1, diagonal = 1, identity = 1;

    printf("Enter elements of 3x3 matrix:\n");

    for (i = 0; i < 3; i++){
        for (j = 0; j < 3; j++) {
            scanf("%d", &a[i][j]);
        }
    }
    for (i = 0; i < 3; i++){
        for (j = 0; j < 3; j++) {
            if (a[i][j] != 0)
                null = 0;

            if (i != j && a[i][j] != 0)
                diagonal = 0;

            if (i == j && a[i][j] != 1)
                identity = 0;

            if (i != j && a[i][j] != 0)
                identity = 0;
        }
    }
    printf("\nSquare Matrix\n");

    if (null == 1)
        printf("Null Matrix\n");

    if (diagonal == 1)
        printf("Diagonal Matrix\n");

    if (identity == 1)
        printf("Identity Matrix\n");

    printf("\nDiagonal Elements: ");
    for (i = 0; i < 3; i++)  {
        printf("%d ", a[i][i]);
    }
    printf("\nUpper Triangular Elements:\n");
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++){
            if (i <= j)
                printf("%d ", a[i][j]);
        }
        printf("\n");
    }
    printf("\nLower Triangular Elements:\n");
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++){
            if (i >= j)
                printf("%d ", a[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
Enter elements of 3x3 matrix:
1  2  3
3  1  4
5  7  1

Square Matrix

Diagonal Elements: 1 1 1
Upper Triangular Elements:
1 2 3
1 4
1

Lower Triangular Elements:
1
3 1
5 7 1
```